

MODULE-1

Computer network -- A computer network is a system that connects two or more devices—like computers, printers, and servers—to share data, resources, and services over physical cables or wireless connections.

INTRODUCTION TO DATA COMMUNICATION--



Data Communication--

Data communication refers to the process of transmitting and receiving data between two or more devices over a communication channel. It involves the conversion of data into signals that can be transmitted and then decoding those signals at the receiving end. Effective data communication requires the use of appropriate protocols, encoding techniques, and hardware devices.

Components of Data Communication--

Data communication systems consist of several components that work together to enable the transfer of data. These components include:

- **Sender:** The device or system that initiates the data transmission.
- **Receiver:** The device or system that receives the transmitted data.
- **Medium/Channel:** The physical pathway through which data is transmitted, such as cables or wireless connections.
- **Protocol:** A set of rules and conventions that govern the transmission and reception of data.
- **Modem:** Short for modulator-demodulator, a device that converts digital signals into analog signals for transmission and vice versa.

Types of Data Transmission--

Data transmission can occur in two primary ways:

Serial Transmission: In serial transmission, data is transmitted bit by bit over a single communication channel. It is commonly used for long-distance communication and is more reliable but slower compared to parallel transmission.

Parallel Transmission: In parallel transmission, multiple bits are transmitted simultaneously over separate communication channels. It allows for faster data transfer but is more susceptible to errors in long-distance transmissions.

Network application—

A **network application** is a software program which operates over a network and allows communication and data sharing between multiple devices and users. Various essential technologies, often referred to as building blocks, serve as foundational elements that developers can utilize to create scalable and effective applications within a networking infrastructure.

There is a wide range of network applications that are used on different devices like smartphones, tablets, laptops, servers, etc. In fact, every device connected to a network relies on some form of network application to communicate effectively within that network.

Types of network applications

Network applications come in various forms, each serving different purposes and use cases. Here are some common types:

- **Client-server applications**

These applications involve a client requesting services or resources from a server. The server then provides the requested information or performs the necessary actions. Examples include web browsers accessing web servers and email clients communicating with mail servers.

- **Peer-to-peer applications**

Unlike client-server applications, peer-to-peer (P2P) applications enable direct communication between devices without relying on a central server. Popular examples include file-sharing programs like BitTorrent and communication tools like Skype.

- **Real-time applications** These applications require low latency and high-speed data transfer to function effectively. Examples include video conferencing tools like Zoom and online gaming platforms, where real-time interaction is crucial.

- **Batch applications** Batch applications process data in batches rather than in real-time. They are often used for tasks such as data backup, file transfer, and large-scale data processing. Examples include automated backup systems and batch file transfer utilities.

Network Hardware—

Network hardware is a set of physical or network devices that are essential for interaction and communication between hardware units operational on a computer network. These are dedicated hardware components that connect to each other and enable a network to function effectively and efficiently.

Network hardware plays a key role as industries grow as it supports scalability. It integrates any number of components depending on the enterprise's needs. Network hardware helps establish an effective mode of communication, thereby improving the business standards. It also promotes multiprocessing and enables sharing of resources, information, and software with ease.

Network equipment is part of advancements of the Ethernet network protocol and utilizes a twisted pair or fiber cable as a connection medium. Routers, hubs, switches, and bridges are some examples of network hardware.

Let's look at the fundamental devices of a computer network.

- **Modems:** A modem enables a computer to connect to the internet via a telephone line. The modem at one end converts the computer's digital signals into analog signals and sends them through a telephone line. At the other end, it converts the analog signals to digital signals that are understandable for another computer.
- **Routers:** A router connects two or more networks. One common use of the router is to connect a home or office network (LAN) to the internet (WAN). It generally has a plugged-in internet cable along with cables that connect computers on the LAN. Alternatively, a LAN connection can also be wireless (Wi-Fi-enabled), making the network device wireless. These are also referred to as wireless access points (WAPs).
- **Hubs, bridges, and switches:** Hubs, bridges, and switches are connecting units that allow multiple devices to connect to the router and enable data transfer to all devices on a network. A router is a complex device with the capabilities of hubs, bridges, and even switches.

Hubs: A hub broadcasts data to all devices on a network. As a result, it consumes a lot of bandwidth as many computers might not need to receive the broadcasted data. The hub could be useful in linking a few gaming consoles in a local multiplayer game via a wired or wireless LAN.

Bridges: A bridge connects two separate LAN networks. It scans for the receiving device before sending a message. This implies that it avoids unnecessary data transfers if the receiving device is not there. Moreover, it also checks to see whether the receiving device has already received the message. These practices improve the overall performance of the network.

Switches: A switch is more powerful than a hub or a bridge but performs a similar role. It stores the MAC addresses of network devices and transfers data packets only to those devices that have requested. Thus, when the demand is high, a switch becomes more efficient as it reduces the amount of latency.

- **Network interface cards:** A network interface card (NIC) is a hardware unit installed on a computer, which allows it to connect to a network. It is typically in the form of a circuit board or chip. In most modern machines, NICs are built into the motherboards, while in some computers, an extra expansion card in the form of a small circuit board is added externally.
- **Network cables:** Cables connect different devices on a network. Today, most networks have cables over a wireless connection as they are more secure, i.e., less prone to attacks, and at the same time carry larger volumes of data per second.

- **Firewall:** A firewall is a hardware or software device between a computer and the rest of the network open to attackers or hackers. Thus, a LAN can be protected from hackers by placing a firewall between the LAN and the internet connection. A firewall allows authorized connections and data-like emails or web pages to pass through but blocks unauthorized connections made to a computer or LAN.

Protocol—

A protocol is a set of rules that determines how data is sent and received over a network. The protocol is just like a language that computers use to talk to each other, ensuring they understand and can respond to each other's messages correctly. Protocols help make sure that data moves smoothly and securely between devices on a network.

To make communication successful between devices, some rules and procedures should be agreed upon at the sending and receiving ends of the system. Such rules and procedures are called Protocols. Different types of protocols are used for different types of communication.



Protocols

In the above diagrams, Protocols are shown as a set of rules. Communication between the Sender and Receiver is not possible without Protocol.

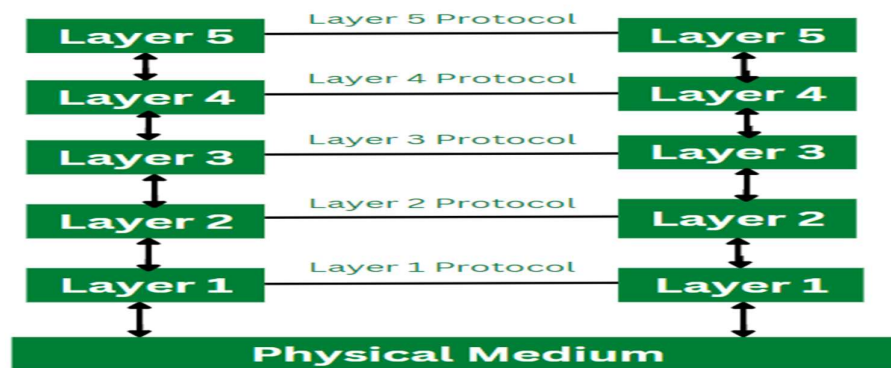
Key Elements of Protocol

- **Syntax:** Syntax refers to the structure or the format of the data that gets exchanged between the devices. Syntax of the message includes the type of data, composition of the message, and sequencing of the message. The starting 8 bits of data are considered as the address of the sender. The next 8 bits are considered to be the address of the receiver. The remaining bits are considered as the message itself.
- **Semantics:** Semantics defines data transmitted between devices. It provides rules and norms for understanding message or data element values and actions.
- **Timing:** Timing refers to the synchronization and coordination between devices while transferring the data. Timing ensures at what time data should be sent and how fast data can be sent. For example, If a sender sends 100 Mbps but the receiver can only handle 1 Mbps, the receiver will overflow and lose data. Timing ensures preventing of data loss, collisions, and other timing-related issues.
- **Sequence Control:** Sequence control ensures the proper ordering of data packets. The main responsibility of sequence control is to acknowledge the data while it get received, and the retransmission of lost data. Through this mechanism, the data is delivered in correct order.

- **Flow Control:** Flow control regulates device data delivery. It limits the sender's data or asks the receiver if it's ready for more. Flow control prevents data congestion and loss.
- **Error Control:** Error control mechanisms detect and fix data transmission faults. They include error detection codes, data resend, and error recovery. Error control detects and corrects noise, interference, and other problems to maintain data integrity.
- **Security :** Network security protects data confidentiality, integrity, and authenticity. which includes encryption, authentication, access control, and other security procedures. Network communication's privacy and trustworthiness are protected by security standards.

Introduction to Layered Architecture--

Every network consists of a specific number of functions, layers, and tasks to perform. Layered Architecture in a computer network is defined as a model where a whole network process is divided into various smaller sub-tasks. These divided sub-tasks are then assigned to a specific layer to perform only the dedicated tasks. A single layer performs only a specific type of task. To run the application and provide all types of services to clients a lower layer adds its services to the higher layer present above it. Therefore layered architecture provides interactions between the sub-systems. If any type of modification is done in one layer it does not affect the next layer.



Layered Architecture

As shown in the above diagram, there are five different layers. Therefore, it is a five-layered architecture. Each layer performs a dedicated task. The lower-level data for example from layer 1 data is transferred to layer 2. Below all the layers Physical Medium is present. The physical medium is responsible for the actual communication to take place. For the transfer of data and communication layered architecture provides simple interface.

Features of Layered Architecture--

- Use of Layered architecture in computer network provides with the feature of modularity and distinct interfaces.
- Layered architecture ensures independence between layers, by offering services to higher layers from the lower layers and without specifying how these services are implemented.
- Layered architecture segments as larger and unmanageable design into small sub tasks.

- In layer architecture every network has different number of functions, layers and content.
- In layered architecture, the physical route provides with communication which is available under the layer 1.
- In layered architecture, the implementation done by one layer can be modified by another layer.

Elements of Layered Architecture

There are three different types of elements of a layered architecture. They are described below:

- **Service:** Service is defined as a set of functions and tasks being provided by a lower layer to a higher layer. Each layer performs a different type of task. Therefore, actions provided by each layer are different.
- **Protocol:** Protocol is defined as a set rule used by the layer for exchanging and transmission of data with its peer entities. These rules can consists details regarding a type of content and their order passed from one layer to another.
- **Interface:** Interface is defined as a channel that allows to transmit the messages from one layer to the another.

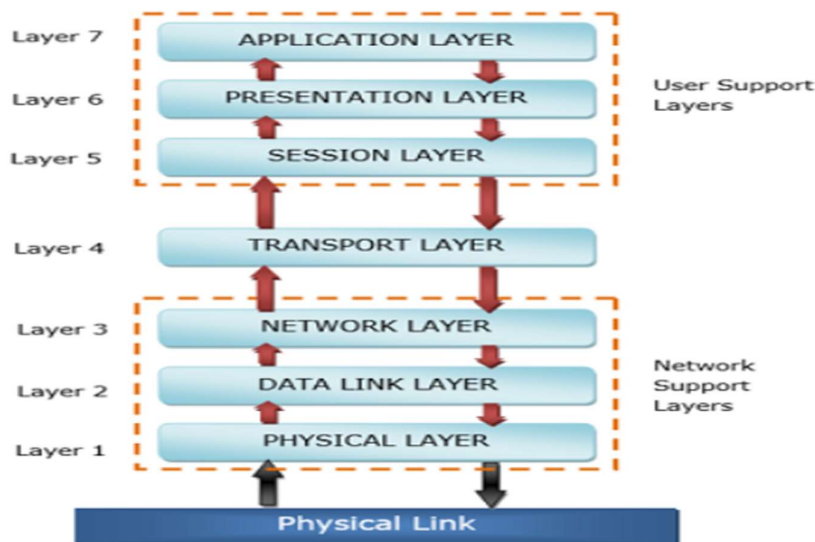
OSI (Open Systems Interconnection) Model—

OSI model or Open Systems Interconnection model serves as a framework that explains the process of information transfer, i.e., from one computer's software application to another computer through a medium. It was created by the International Organization for Standardization (ISO) in 1984. It is now widely acknowledged as an architecture model for inter-computer communication.

OSI model is nothing but an architecture model that assists in breaking down the task of data communication into seven layers. Each layer has its role and further interacts with the adjacent layers. This model works in a way that describes how various network devices and protocols work with one another.

The OSI model is divided into seven layers. These are:

- Application Layer
- Presentation Layer
- Session Layer
- Transport Layer
- Network Layer
- Data Link Layer
- Physical Layer



Inside this model, each layer works on certain network functions and further communicates with both layers, i.e., one above it and the other below it. When we talk about the upper layers (application, presentation, and session), also known as the software layer, we focus on issues related to the applications themselves, such as user interface design, data formatting, and encryption techniques. On the other hand, when we talk about the lower layers (network, data link, and physical), also known as the physical layer, we focus on issues such as error control processes and transmission mediums.

In the middle is the heart of the OSI model, i.e., the transport layer.

Before explaining each layer in detail, let's first understand the purpose of the OSI model.

Purpose of OSI Model

The primary goal of the OSI model is to promote compatibility between network devices and software applications. By adopting the OSI model as a standard reference, network engineers and developers can guarantee collaboration and integration, across platforms and systems.

There are advantages associated with the OSI model. Some of these are:

- It streamlines the network design process by breaking it down into segments.
- It facilitates the development and testing of network components.
- It allows for troubleshooting and problem identification by pinpointing faulty layers.

Let's now discuss each layer in detail.

The 7 Layers of OSI Model

This section will explain the different layers in the OSI model, along with some practical examples.

7. Application Layer

At the top of the OSI model, we have an application layer. This layer generally deals with the end users and serves as a bridge between them and various other applications such as web

browsers, email clients, and file transfer protocols. It further plays a significant role in allowing communication by defining protocols such as HTTP, SMTP, FTP, etc.

Function of Application Layer

Below, we have discussed some of the key functions performed by the application layer.



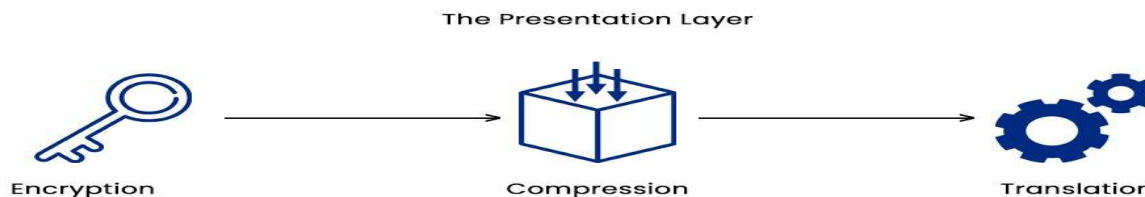
- Identifying communication partners: Here, the application layer assists in determining the parties that are involved in a communication process and further verifies their availability.
- Determining resource availability: It verifies that all the network resources required to communicate are available.
- Synchronizing communication: The application layer is the head that manages and synchronizes interactions between different applications.

6. Presentation Layer

When we talk about the presentation layer, it is simply responsible for translating the data between the application layer that is above it and the lower layers. It also ensures the compatibility format of the data for both the sender as well as receiver applications. Apart from that, it also handles data compression, encryption, and decryption.

Function of Presentation Layer

Below, we have explained some of the main functions of the presentation layer.



- Data Translation: In data translation, the presentation layer simply converts data from one format to another, whichever is suitable for the upper and lower layers. An example is converting ASCII to EBCDIC or binary text.
- Data Compression: When we talk about data compression, the presentation layer reduces the data size to save bandwidth and storage space.
- Data Encryption: In order to maintain the integrity as well as privacy of the data, the presentation layer encrypts it.

5. Session Layer

The session layer assists in establishing, maintaining, and terminating sessions between different applications. It is a logical connection responsible for allowing continuous data

exchange between two applications. Apart from that, it also helps manage authentication, authorization, and synchronization.

Functions of Sessions Layer

Some of the important functions of the session layer are:



- **Session Establishment:** In session establishment, the session layer simply creates sessions between applications by exchanging control information. These include usernames, passwords, and session IDs.
- **Session Maintenance:** The session layer also assists in monitoring and controlling the sessions. This is done by sending acknowledgements, checkpoints, and timeouts.
- **Session termination:** In the end, the session layer ends the session by releasing resources and closing connections.

4. Transport Layer

The transport layer assists in providing reliable and efficient data transfer between two devices within a network. It also ensures that the data that is delivered must be without errors, loss, or duplication. Apart from that, it is also responsible for regulating the flow of data in order to avoid congestion or overload.

TCP and UDP are the most important protocols that are used in the transport layer.

Function of Transport Layer

Below, we have explained some of the functions of the transport layer.



- **Segmentation and Reassembly:** In this case, the transport layer simply divides the data into small segments that can further be transmitted over the network. It also assists in reassembling the segments at the destination device.
- **Error Control:** The transport layer is responsible for detecting and correcting errors in the data with the help of checksums, acknowledgements, and retransmissions.
- **Flow Control:** In the end, the transport layer controls the rate at which the data is transmitted by using windowing, buffering, and many congestion avoidance techniques available.

3. Network Layer

The network layer assists in routing the packets from one device to another in a network. It is also responsible for determining the best path for the packets based on various factors such as distance, cost, and traffic. Apart from that, the network layer also handles addressing, fragmentation, and packet switching.

Function of Network Layer

Now, let's discuss some of the crucial functions of the network Layer.



- **Logical Addressing:** In this, the network layer simply assigns logical addresses, such as IP addresses, to a network device in a network. It also maps logical addresses to physical addresses, such as MAC addresses, using protocols such as ARP and RARP.
- **Routing:** It is the work of the network layer to decide the best path for the packet using algorithms such as Dijkstra's and Bellman-Ford.
- **Fragmentation:** Fragmentation simply means splitting large packets into small segments that can fit the maximum transmission unit (MTU) of the underlying network. Here, the network layer also comes into action.
- **Packet Switching:** The network layer is also responsible for switching the packets from one network to another using devices such as routers and gateways.

Some of the common protocols of the network layer are IP, ICMP, ARP, and RARP.

2. Data Link Layer

The data link layer is used for the node-to-node delivery of data. It further defines the format of data and also assists in providing reliable and efficient communication between two devices. It is also responsible for handling access control, error detection, and correction.

The data link layer is further divided into two sub-layers. These are:

1. **Logical Link Control:** LLC assists in providing services to the upper layers. These services include flow control, error control, and multiplexing.
2. **Media Access Control:** MAC is the one that controls how devices access and share the physical medium, such as Ethernet, Wi-Fi, or Bluetooth.

Functions of Data Link Layer

Now, let's discuss the different functions of the data link layer.



- **Framing:** The data link layer assists in dividing the data into frames that further contain a header, a payload, and a trailer. When we talk about the header, it contains information such as MAC addresses of the source and destination, frame types, and frame numbers. The trailer part contains a checksum for detecting the error.
- **Error Detection:** This layer helps in detecting the errors in the frames by using methods such as parity check, cyclic redundancy check (CRC), and checksum.
- **Error Correction:** The data link layer is also responsible for correcting the errors in the frames by using methods such as stop-and-wait, go-back-N, and selective repeat.
- **Access Control:** The data link layer regulates how devices access and share the physical medium by using methods such as CSMA/CD, CSMA/CA, and token passing.

1. Physical Layer

The last layer of the OSI model is the physical layer. It is closest to the physical medium and is responsible for transmitting as well as receiving individual bits from one device to another. Apart from that, the physical layer specifies the mechanical, electrical, and procedural characteristics of the physical medium.

Functions of Physical Layer

Below, we have explained some of the important functions of the physical layer.



- **Bit Synchronization:** This layer is responsible for the synchronization of bits by providing a clock signal that controls both sender and receiver devices.
- **Bit Rate Control:** It also defines the transmission rate by utilizing various methods. Some of these methods are modulation, encoding, and multiplexing.
- **Physical Topologies:** The physical layer assists in defining how devices are arranged in a network. This includes bus, ring, or mesh topology.
- **Transmission Mode:** One of the most important functions of the physical layer is defining how data flows between devices. This includes simplex, half-duplex, or full-duplex mode.

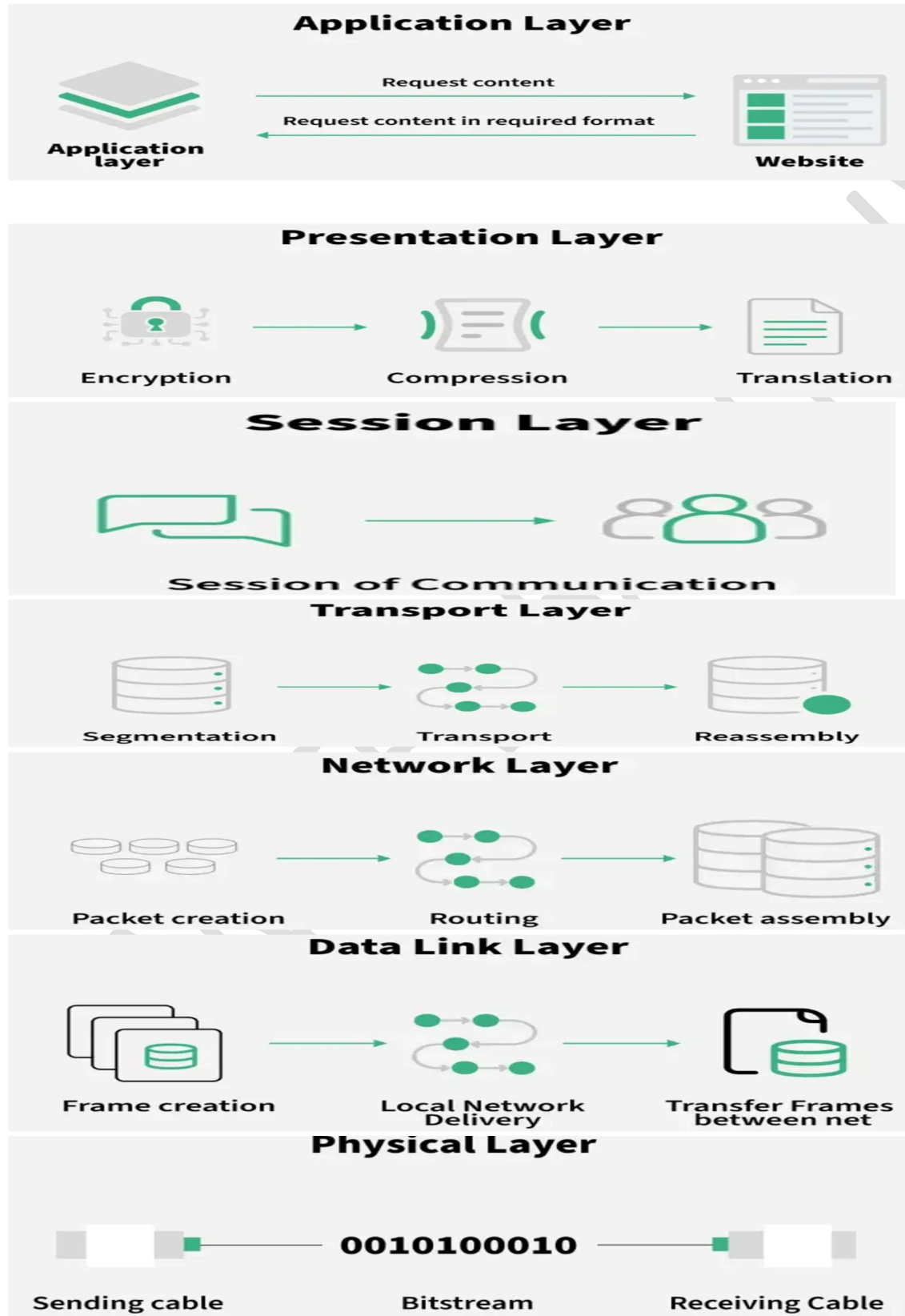
These are the 7 layers of OSI Model.

OR

OSI (Open Systems Interconnection) Model--

The **OSI (Open Systems Interconnection)** Model is a set of rules that explains how different computer systems communicate over a network. OSI Model was developed by the **International Organization for Standardization (ISO)**. The OSI Model consists of 7 layers and each layer has specific functions and responsibilities. This layered approach makes

it easier for different devices and technologies to work together. OSI Model provides a clear structure for data transmission and managing network issues. The OSI Model is widely used as a reference to understand how network systems function.



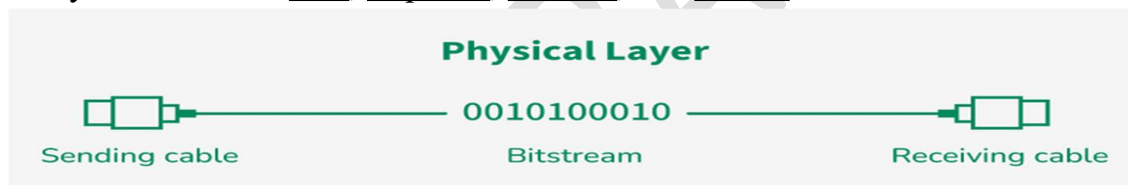
Layers of the OSI Model

There are 7 layers in the OSI Model and each layer has its specific role in handling data. All the layers are mentioned below:

- Physical Layer
- Data Link Layer
- Network Layer
- Transport Layer
- Session Layer
- Presentation Layer
- Application Layer

Layer 1: Physical Layer

The lowest layer of the OSI reference model is the **Physical Layer**. It is responsible for the actual physical connection between the devices. The physical layer contains information in the form of **bits**. Physical Layer is responsible for transmitting individual bits from one node to the next. When receiving data, this layer will get the signal received and convert it into 0s and 1s and send them to the Data Link layer, which will put the frame back together. Common physical layer devices are Hub, Repeater, Modem, and Cables.



Physical Layer

Functions of the Physical Layer

- **Bit Synchronization:** The physical layer provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- **Bit Rate Control:** The Physical layer also defines the transmission rate i.e. the number of bits sent per second.
- **Physical Topologies:** Physical layer specifies how the different, devices/nodes are arranged in a network i.e. bus topology, star topology, or mesh topology.
- **Transmission Mode:** Physical layer also defines how the data flows between the two connected devices. The various transmission modes possible are Simplex, half-duplex and full duplex.

Layer 2: Data Link Layer (DLL)

The data link layer is responsible for the node-to-node delivery of the message. The main function of this layer is to make sure data transfer is error-free from one node to another, over the physical layer. When a packet arrives in a network, it is the responsibility of the DLL to

transmit it to the Host using its MAC address. Packet in the Data Link layer is referred to as Frame. Switches and Bridges are common Data Link Layer devices.

The Data Link Layer is divided into two sublayers:

- Logical Link Control (LLC)
- Media Access Control (MAC)

The packet received from the Network layer is further divided into frames depending on the frame size of the NIC (**Network Interface Card**). DLL also encapsulates Sender and Receiver's MAC address in the header.

The Receiver's MAC address is obtained by placing an ARP (Address Resolution Protocol) request onto the wire asking, "Who has that IP address?" and the destination host will reply with its MAC address.

Functions of the Data Link Layer

- **Framing:** Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver. This can be accomplished by attaching special bit patterns to the beginning and end of the frame.
- **Physical Addressing:** After creating frames, the Data link layer adds physical addresses (MAC addresses) of the sender and/or receiver in the header of each frame.
- **Error Control:** The data link layer provides the mechanism of error control in which it detects and retransmits damaged or lost frames.
- **Flow Control:** The data rate must be constant on both sides else the data may get corrupted thus, flow control coordinates the amount of data that can be sent before receiving an acknowledgment.
- **Access Control:** When a single communication channel is shared by multiple devices, the MAC sub-layer of the data link layer helps to determine which device has control over the channel at a given time.

Layer 3: Network Layer

The network layer works for the transmission of data from one host to the other located in different networks. It also takes care of packet routing i.e. selection of the shortest path to transmit the packet, from the number of routes available. The sender and receiver's IP address are placed in the header by the network layer. Segment in the Network layer is referred to as Packet. Network layer is implemented by networking devices such as routers and switches.

Functions of the Network Layer

- **Routing:** The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing.
- **Logical Addressing:** To identify each device inter-network uniquely, the network layer defines an addressing scheme. The sender and receiver's IP addresses are placed in the

header by the network layer. Such an address distinguishes each device uniquely and universally.

Layer 4: Transport Layer

The transport layer provides services to the application layer and takes services from the network layer. The data in the transport layer is referred to as **Segments**. It is responsible for the end-to-end delivery of the complete message. The transport layer also provides the acknowledgment of the successful data transmission and re-transmits the data if an error is found. Protocols used in Transport Layer are TCP, UDP, NetBIOS, PPTP.

At the sender's side, the transport layer receives the formatted data from the upper layers, performs **Segmentation**, and also implements **Flow and error control** to ensure proper data transmission. It also adds Source and Destination port number in its header and forwards the segmented data to the Network Layer.

- Generally, this destination port number is configured, either by default or manually. For example, when a web application requests a web server, it typically uses port number 80, because this is the default port assigned to web applications. Many applications have default ports assigned.

At the Receiver's side, Transport Layer reads the port number from its header and forwards the Data which it has received to the respective application. It also performs sequencing and reassembling of the segmented data.

Functions of the Transport Layer

- **Segmentation and Reassembly:** This layer accepts the message from the (session) layer and breaks the message into smaller units. Each of the segments produced has a header associated with it. The transport layer at the destination station reassembles the message.
- **Service Point Addressing:** To deliver the message to the correct process, the transport layer header includes a type of address called service point address or port address. Thus, by specifying this address, the transport layer makes sure that the message is delivered to the correct process.

Services Provided by Transport Layer

- Connection-Oriented Service
- Connectionless Service

Layer 5: Session Layer

Session Layer in the OSI Model is responsible for the establishment of connections, management of connections, terminations of sessions between two devices. It also provides authentication and security. Protocols used in the Session Layer are NetBIOS, PPTP.

Functions of the Session Layer

- **Session Establishment, Maintenance, and Termination:** The layer allows the two processes to establish, use, and terminate a connection.

- **Synchronization:** This layer allows a process to add checkpoints that are considered synchronization points in the data. These synchronization points help to identify the error so that the data is re-synchronized properly, and ends of the messages are not cut prematurely, and data loss is avoided.
- **Dialog Controller:** The session layer allows two systems to start communication with each other in half-duplex or full duplex.



Communication in Session Layer

Layer 6: Presentation Layer

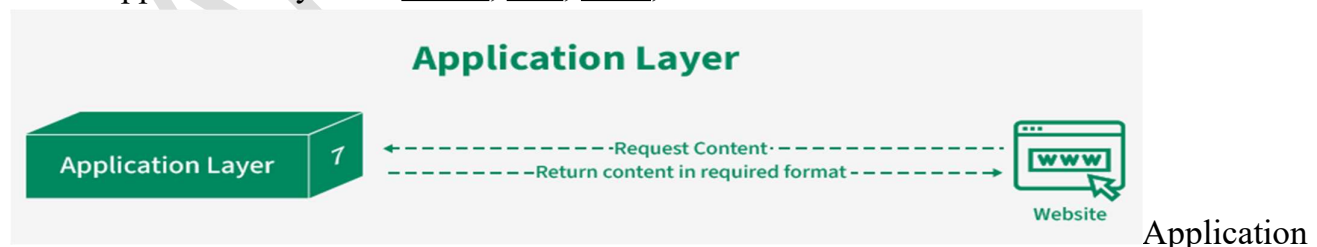
The presentation layer is also called the **Translation layer**. The data from the application layer is extracted here and manipulated as per the required format to transmit over the network. Protocols used in the Presentation Layer are TLS/SSL (Transport Layer Security / Secure Sockets Layer). JPEG, MPEG, GIF, are standards or formats used for encoding data, which is part of the presentation layer's role.

Functions of the Presentation Layer

- **Translation:** For example, ASCII to EBCDIC.
- **Encryption/ Decryption:** Data encryption translates the data into another form or code. The encrypted data is known as the ciphertext, and the decrypted data is known as plain text. A key value is used for encrypting as well as decrypting data.
- **Compression:** Reduces the number of bits that need to be transmitted on the network.

Layer 7: Application Layer

At the very top of the OSI Reference Model stack of layers, we find the Application layer which is implemented by the network applications. These applications produce the data to be transferred over the network. This layer also serves as a window for the application services to access the network and for displaying the received information to the user. Protocols used in the Application layer are SMTP, FTP, DNS, etc.



Layer

Functions of the Application Layer

The main functions of the application layer are given below.

- **Network Virtual Terminal (NVT):** It allows a user to log on to a remote host.

- **File Transfer Access and Management (FTAM):** This application allows a user to access files in a remote host, retrieve files in a remote host, and manage or control files from a remote computer.
- **Mail Services:** Provide email service.
- **Directory Services:** This application provides distributed database sources and access for global information about various objects and services.

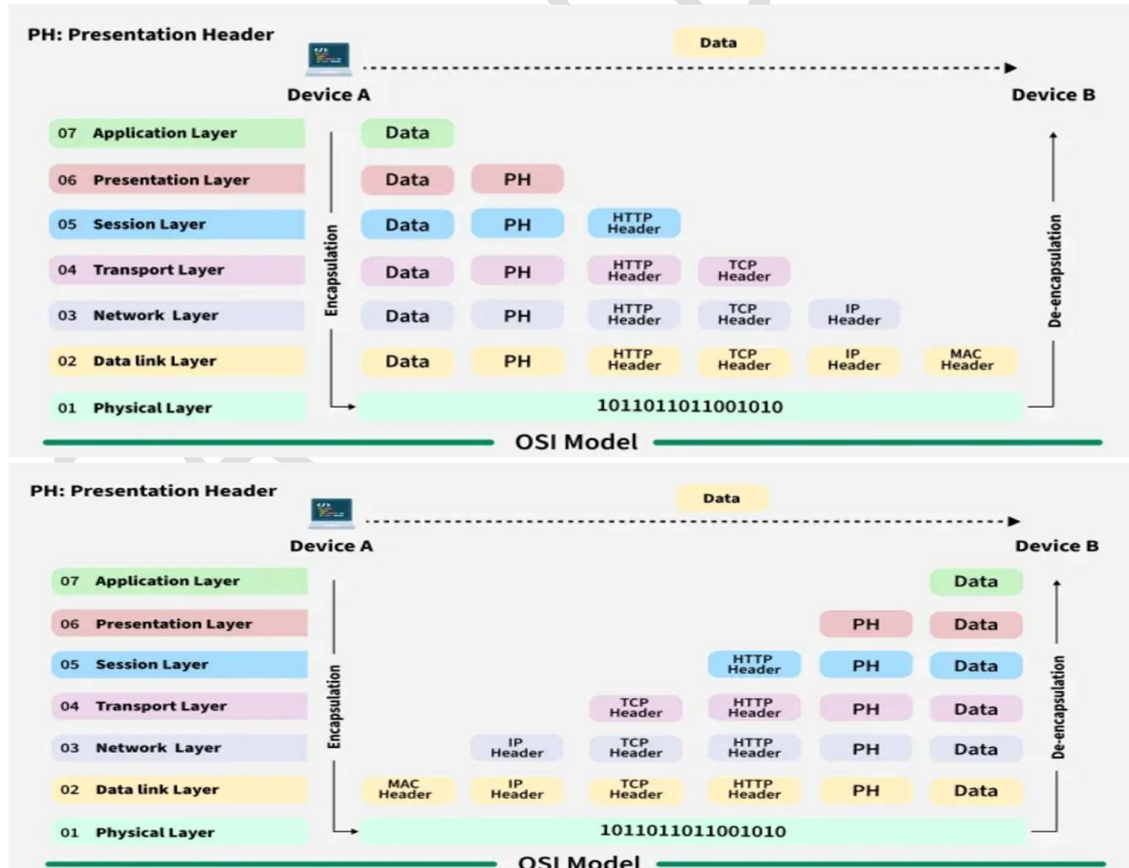
Data Flows in the OSI Model--

When we transfer information from one device to another, it travels through 7 layers of OSI model. First data travels down through 7 layers from the sender's end and then climbs back 7 layers on the receiver's end.

Data flows through the OSI model in a step-by-step process:

- **Application Layer:** Applications create the data.
- **Presentation Layer:** Data is formatted and encrypted.
- **Session Layer:** Connections are established and managed.
- **Transport Layer:** Data is broken into segments for reliable delivery.
- **Network Layer:** Segments are packaged into packets and routed.
- **Data Link Layer:** Packets are framed and sent to the next device.
- **Physical Layer:** Frames are converted into bits and transmitted physically.

Each layer adds specific information to ensure the data reaches its destination correctly, and these steps are reversed upon arrival.



We can understand how data flows through OSI Model with the help of an example mentioned below.

Let us suppose, **Person A** sends an e-mail to his friend **Person B**.

Step 1: Person A interacts with e-mail application like **Gmail, outlook**, etc. Writes his email to send. (This happens at **Application Layer**).

Step 2: At Presentation Layer, Mail application prepares for data transmission like encrypting data and formatting it for transmission.

Step 3: At Session Layer, there is a connection established between the sender and receiver on the internet.

Step 4: At Transport Layer, Email data is broken into smaller segments. It adds sequence number and error-checking information to maintain the reliability of the information.

Step 5: At Network Layer, addressing of packets is done in order to find the best route for transfer.

Step 6: At Data Link Layer, data packets are encapsulated into frames, then MAC address is added for local devices and then it checks for error using error detection.

Step 7: At Physical Layer, Frames are transmitted in the form of electrical/ optical signals over a physical network medium like ethernet cable or WiFi.

Protocols Used in the OSI Layers

Layer	Working	Protocol Data Unit	Protocols
1: Physical Layer	Establishing Physical Connections between Devices.	Bits	<u>USB</u> , <u>SONET/SDH</u> , etc.
2: Data Link Layer	Node to Node Delivery of Message.	Frames	<u>Ethernet</u> , PPP, etc.

Layer	Working	Protocol Data Unit	Protocols
3: Network Layer	Transmission of data from one host to another, located in different networks.	Packets	IP, <u>ICMP</u> , <u>IGMP</u> , <u>OSPF</u> , etc.
4: Transport Layer	Take Service from Network Layer and provide it to the Application Layer.	Segments (for TCP) or Datagrams (for UDP)	<u>TCP</u> , <u>UDP</u> , <u>SCTP</u> , etc.
5: Session Layer	Establishes Connection, Maintenance, Ensures Authentication and Ensures security.	Data	<u>NetBIOS</u> , <u>RPC</u> , <u>PPTP</u> , etc.
6: Presentation Layer	Data from the application layer is extracted and manipulated in the required format for transmission.	Data	<u>TLS/SSL</u> , <u>MIME</u> , etc.
7: Application Layer	Helps in identifying the client and synchronizing communication.	Data	<u>FTP</u> , <u>SMTP</u> , <u>DNS</u> , <u>DHCP</u> , etc.

The OSI Model defines the communication of a computing system into 7 different layers. Its advantages include:

- It divides network communication into 7 layers which makes it easier to understand and troubleshoot.
- It standardizes network communications, as each layer has fixed functions and protocols.
- Diagnosing network problems is easier with the **OSI model**.

- It is easier to improve with advancements as each layer can get updates separately.

Disadvantages of OSI Model

- The OSI Model has seven layers, which can be complicated and hard to understand for beginners.
- In real-life networking, most systems use a simpler model called the Internet protocol suite (TCP/IP), so the OSI Model is not always directly applicable.
- Each layer in the OSI Model adds its own set of rules and operations, which can make the process more time-consuming and less efficient.
- The OSI Model is more of a theoretical framework, meaning it's great for understanding concepts but not always practical for implementation.

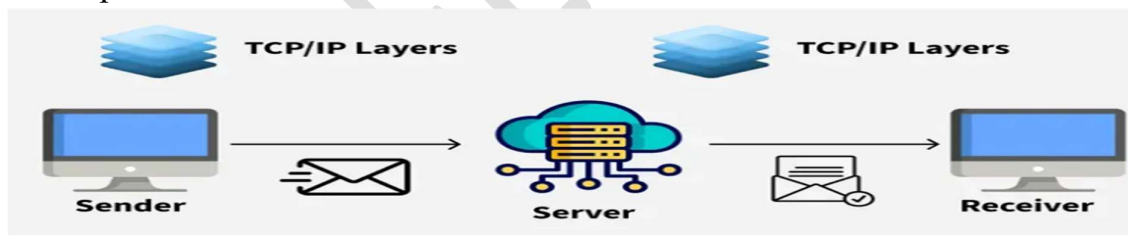
The TCP/IP model ---

The TCP/IP model is a framework that is used to model the communication in a network. It is mainly a collection of network protocols and organization of these protocols in different layers for modeling the network.

- It has four layers, Application, Transport, Network/Internet and Network Access.
- While the OSI model has seven layers, the 4 layer TCP/IP model is simpler and commonly used in today's Internet and networking systems.

Role of TCP/IP

One of its main goals is to make sure that the data sent by the sender arrives safely and correctly at the receiver's end. To do this, the data is broken down into smaller parts called packets before being sent. These packets travel separately and are reassembled in the correct order when they reach the destination. This helps prevent errors and makes sure the message is complete and accurate.



TCP

Layers of TCP/IP Model

1. Application Layer

The Application Layer is the top layer of the TCP/IP model and the one closest to the user. This is where all the apps you use like web browsers, email clients, or file sharing tools connect to the network. It acts like a bridge between your software (like Chrome, Gmail, or WhatsApp) and the lower layers of the network that actually send and receive data.

It supports different protocols like HTTP (for websites), FTP (for file transfers), SMTP (for emails), and DNS (for finding website addresses). It also manages things like data formatting,

so both sender and receiver understand the data, encryption to keep data safe, and session management to keep track of ongoing connections.

2. Transport Layer

The Transport Layer is responsible for making sure that data is sent reliably and in the correct order between devices. It checks that the data you send like a message, file, or video arrives safely and completely. This layer uses two main protocols: TCP and UDP, depending on whether the communication needs to be reliable or faster.

TCP is used when data must be correct and complete, like when loading a web page or downloading a file. It checks for errors, resends missing pieces, and keeps everything in order. On the other hand, UDP (User Datagram Protocol) is faster but doesn't guarantee delivery useful for things like live video or online games where speed matters more than perfect accuracy.

3. Internet Layer

The Internet Layer is used for finding the best path for data to travel across different networks so it can reach the right destination. It works like a traffic controller, helping data packets move from one network to another until they reach the correct device. This layer uses the Internet Protocol (IP) to give every device a unique IP address, which helps identify where data should go.

The main job of this layer is routing deciding the best way for data to travel. It also takes care of packet forwarding (moving data from one point to another), fragmentation (breaking large data into smaller parts), and addressing.

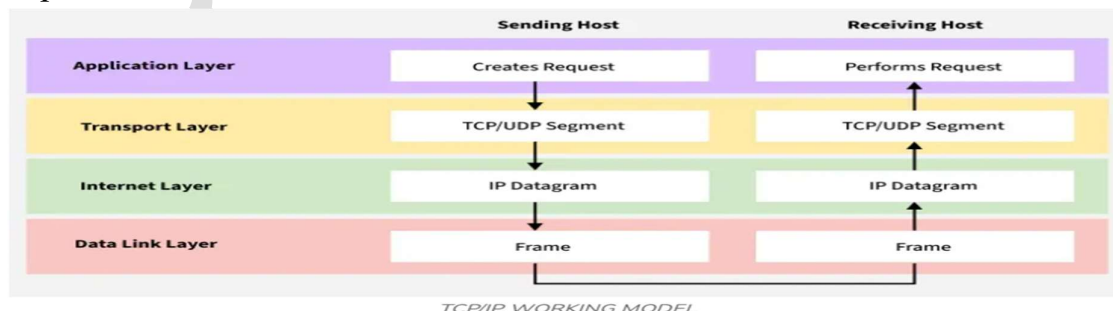
4. Network Access Layer

The Network Access Layer is the bottom layer of the TCP/IP model. It deals with the actual physical connection between devices on the same local network like computers connected by cables or communicating through Wi-Fi. This layer makes sure that data can travel over the hardware, such as wires, switches, or wireless signals.

It also handles important tasks like using MAC addresses to identify devices, creating frames (the format used to send data over the physical link), and checking for basic errors during transmission.

Working of TCP/IP Model

The working of TCP/IP can be explained with the help of the diagram given below and explained:



When Sending Data (From Sender to Receiver)

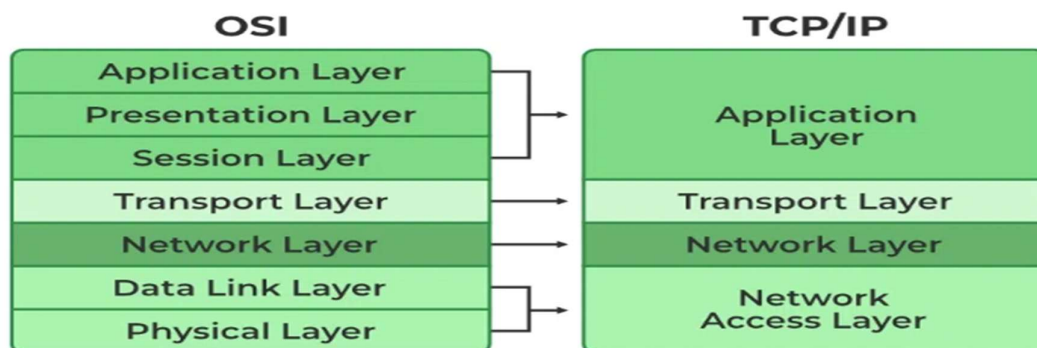
- **Application Layer:** Prepares user data using protocols like HTTP, FTP, or SMTP.
- **Transport Layer (TCP/UDP):** Breaks data into segments and ensures reliable (TCP) or fast (UDP) delivery.
- **Internet Layer (IP):** Adds IP addresses and decides the best route for each packet.
- **Link Layer (Network Access Layer):** Converts packets into frames and sends them over the physical network.

When Receiving Data (At the Destination)

- **Link Layer:** Receives bits from the network and rebuilds frames to pass to the next layer.
- **Internet Layer:** Checks the IP address, removes the IP header, and forwards data to the Transport Layer.
- **Transport Layer:** Reassembles segments, checks for errors, and ensures data is complete.
- **Application Layer:** Delivers the final data to the correct application (e.g., displays a web page in the browser).

Why TCP/IP is Used Over the OSI Model--

TCP/IP is used over the OSI model because it is simpler, practical, and widely adopted for real-world networking and the internet. The diagram below shows the comparison of OSI layer with the TCP :



OSI

Reason	Explanation
Simpler Structure	TCP/IP has only 4 layers, compared to 7 in OSI, making it easier to implement and understand in real systems.
Protocol-Driven Design	TCP/IP was designed based on working protocols, while the OSI model is more of a theoretical framework.

Reason	Explanation
Flexibility and Robustness	TCP/IP adapts well to different hardware and networks and includes error handling, routing, and congestion control.
Open Standard	TCP/IP is open, free to use, and not controlled by any single organization, helping it gain universal acceptance.
Actual Use vs Conceptual Model	The OSI model is great for education and design principles, but TCP/IP is the one actually used in real-world networking.

Advantages of TCP/IP Model

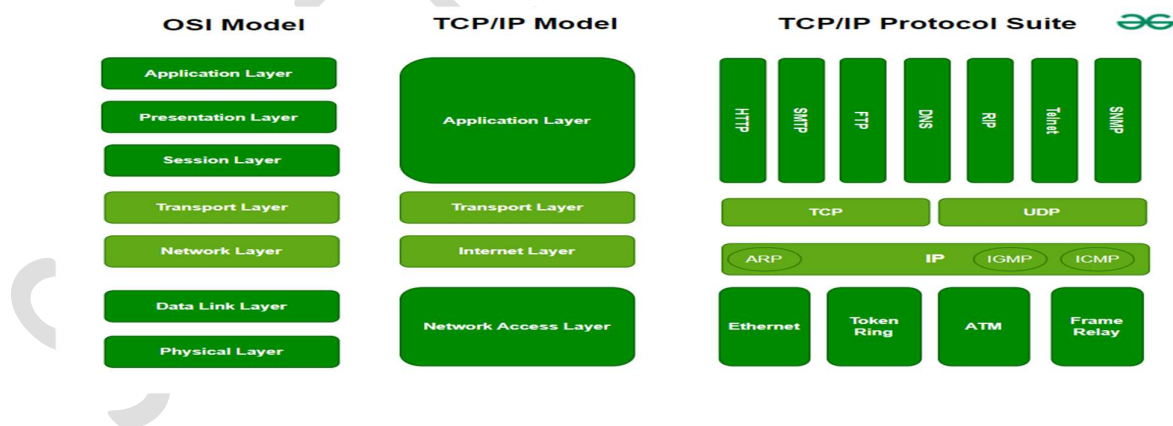
- **Interoperability** : The TCP/IP model allows different types of computers and networks to communicate with each other, promoting compatibility and cooperation among diverse systems.
- **Scalability** : TCP/IP is highly scalable, making it suitable for both small and large networks, from local area networks (LANs) to wide area networks (WANs) like the internet.
- **Standardization** : It is based on open standards and protocols, ensuring that different devices and software can work together without compatibility issues.
- **Flexibility** : The model supports various routing protocols, data types, and communication methods, making it adaptable to different networking needs.
- **Reliability** : TCP/IP includes error-checking and retransmission features that ensure reliable data transfer, even over long distances and through various network conditions.

Disadvantages of TCP/IP Model

- **Security Concerns** : TCP/IP was not originally designed with security in mind. While there are now many security protocols available (such as SSL/TLS), they have been added on top of the basic TCP/IP model, which can lead to vulnerabilities.
- **Inefficiency for Small Networks** : For very small networks, the overhead and complexity of the TCP/IP model may be unnecessary and inefficient compared to simpler networking protocols.
- **Limited by Address Space** : Although IPv6 addresses this issue, the older IPv4 system has a limited address space, which can lead to issues with address exhaustion in larger networks.
- **Data Overhead** : TCP the transport protocol, includes a significant amount of overhead to ensure reliable transmission.

Difference Between OSI and TCP/IP Model

OSI Model	TCP/IP Model
OSI stands for Open Systems Interconnection.	TCP/IP stands for Transmission Control Protocol/Internet Protocol.
OSI model has 7 layers.	TCP/IP model consists of 4 layers.
Package delivery is guaranteed in OSI Model.	Package delivery is not guaranteed in the TCP/IP Model.
In the OSI model, only layers 1,2 and 3 are necessary for data transmission.	All layers of the TCP/IP model are needed for data transmission.
Protocols at each layer is independent of the other layer.	Layers are integrated; some layers are required by other layers of TCP/IP model.
OSI Model is a conceptual framework, less used in practical applications.	Widely used in actual networks like Internet and Communication Systems.



Internet History--

The **internet** as we know it today is a vast, interconnected network that has transformed how we communicate, work, learn, and interact. But it wasn't always this way. The origin of the internet dates back to the 1950s when the world was grappling with the Cold War, and researchers sought to create a communication system that could withstand disruptions. From

its humble beginnings as a military project to the global web it is now, the **history of the internet** is a tale of human ingenuity, collaboration, and the relentless pursuit of connectivity and a fascinating journey of innovation, collaboration, and rapid technological advancement.

Key Milestones in Internet History

- **1969:** Launch of **ARPANET**, the first operational packet-switching network.
- **1971:** The first **email** is sent by **Ray Tomlinson**.
- **1983:** Adoption of **TCP/IP** as the standard protocol for ARPANET.
- **1989-1990:** **Tim Berners-Lee** invents the **World Wide Web**.
- **1993:** The first graphical web browser, **Mosaic**, is released.
- **1998:** **Google** is founded, revolutionizing search engines.
- **2007:** The **iPhone** introduces widespread mobile Internet access.
- **2010s:** Rise of **cloud computing**, **social media**, and **streaming services**.

Vint Cerf and Bob Kahn

Known as

"The Fathers of the Internet"

Key Contributions:

- *Invented TCP/IP protocol in 1974*
- *Created the foundation for data transmission*
- *Developed internet architecture*
- *Established basic internet communication standards*

Internet standards are fundamental to the global internet, which given the complex and diverse nature of the systems and technologies on the internet attempts to prove how these systems can interact. These are standardization patterns that are provided by the Internet Engineering Task Force (IETF), which has worked on its protocols for a long time and has a strict mechanism for revising and approving them.

These standards will facilitate web affinity to signal, security, and innovation. In this article, we will look into the standardization process in Internet Standard, its importance, examples, etc.

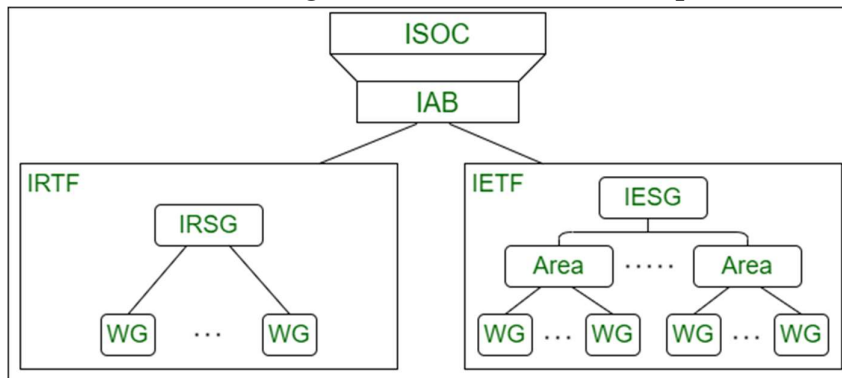
Primary Technologies

- **Internet Standard(STD):** proposed as its formal specification of the rule and policy promoting the uniform and compatibility usage of Internet architecture across the world.
- **Request For Comments (RFC):** The entire collection of documents that formed the foundation for developing the internet standards and protocols and also for their promotion and implementation.
- **Proposed Standard:** It is used at a certain level of the work in order to prove that the current specification is not just prepared by somebody and exists in the form of a

planned work, but has gone through the basic level of evaluation and the review and standardization phase of this work.

- **Draft Standard:** A better-defined progression of a concept that is a previously rational and developed idea that has been implemented in other settings.
- **Internet Standard:** A frequently adopted and highly stable standard that has turned out to be very effective in actual practice as far as interconnect is concerned.

Internet Administration is basically a group that coordinates and guides the Internet with its growth and development. It makes sure that all the protocols are followed by the devices and network for the smooth functioning of the internetwork. Some of the organizations that overlook the growth and development of the internet are -



Internet Society (ISOC) : An international, non-profit organization, ISOC formed to provide support for the Internet standard process accomplishes its goals by maintaining and supporting other Internet administrative bodies such as IAB, IETF, IRTF, and IANA. It also promotes research and other activities relating to the Internet. **Internet Architecture Board (IAB) :** The technical advisor to the ISOC, the IAB's main purpose is to oversee the continuing development of the TCP/IP Protocol Suite and to serve in a technical advisory capacity to research members of the Internet community, which is accomplished by its two components, the Internet Engineering Task Force (IETF) and the Internet Research Task Force (IRTF). Additionally, IAB is also the editorial manager of the RFCs and is the external liaison between the Internet and the other standards, organizations, and forums. **Internet Engineering Task Force (IETF) :** Managed by the Internet Engineering Steering Group (IESG), the IETF is a forum of working groups responsible for identifying operational problems, proposing solutions to them, and developing and reviewing specifications intended as Internet Standards. The groups are collected into areas, like, applications, routing, security, protocols, and network management, and each area concentrates on a specific topic. **Internet Research Task Force (IRTF) :** Managed by the Internet Research Steering Group (IRSG), the IRTF is a forum of working groups focusing on long-term research topics related to Internet protocol, technology, applications, and architecture.